Trainings

General Information

Automotive gasolines are used to fuel internal combustion spark-ignition engines. Gasoline fuel is a blend of numerous hydrocarbons derived from petroleum. To produce gasoline fuel, refiners initially use fractional distillation of the crude oil to segregate those hydrocarbons in the gasoline fuel boiling range, with finished gasoline encompassing a boiling range of about 30 to 225°C [89 to 437°F].

The quality of gasoline fuel is controlled by various industry standards according to the region where the fuel is used. See Table 1 for an abbreviated list of countries and the appropriate specifications:

Table 1, Gasoline Fuel Specification by Country		
Country	Specification	Description
United States of America	ASTM D4814	Gasoline and ethanol blends
Canada	CAN/CGSB-3.5	Gasoline
	CAN/CGSB-3.511	Gasoline and ethanol blends

Specifications

Gasoline fuels that meet national and international specifications may be used in Cummins engines, provided they also comply with the requirements outlined in Table 2 of this document. In the event of conflicting requirements, those specified in this document will be considered the minimum standard. Additionally, the fuel should meet the volatility class and vapor lock protection requirements defined in ASTM D4814 Table 1, based on the seasonal and geographic conditions where it is used. Volatility and vapor lock protection codes are generally **not** known to retail customers, as these are requirements managed by upstream fuel producers and distributors.

Table 2, Cummins Required Fuel Properties			
Property	Minimum	Maximum	Test Method
Volatility and Vapor Lock Protection class	At point of use fuel must meet appropriate Volatility & Vapor Lock Protection classes based on seasonal, and geographic schedule per ASTM D4814 Table 4		D4953, D5191, D5482, D6378, D86, D7345, D5188

Table 2, Cummins Required Fuel Properties			
Antiknock Index (RON+MON)/2, Index Number	87		ASTM D2699 – Research Octane Number ASTM D2700 – Motor Octane Number
Ethanol Content[1], % by volume		15 (See Note 1)	ASTM D5501
Water content, visual		No visible water or phase separation	ASTM D4176
Particulate Contamination, visual		No visible particulate matter	ASTM D4176
Sulfur[2], mg/kg		10	ASTM D5453
Trace contaminant elements[3], mg/kg	Limit for each element (metals, non-metal, and metalloids) should be <0.5		ASTM D7111

1. Ethanol blends in gasoline composing of 15-30 percent by volume can be utilized in Cummins engines but are **not** recommended. Ethanol-gasoline fuel blends less than 30 percent are **not** suitable for use in Cummins engines.
2. According to ASTM D4814, fuel producers **must** maintain an annual average sulfur concentration in gasoline of 10 ppm or less. However, the specification allows for downstream samples to contain up to 95 ppm sulfur on a single spot-check basis.
3. Trace contaminant elements are undesirable and are **not** intended to be present in the fuel as originally produced. Some contaminants, such as iron (Fe), silicon (Si), and zinc (Zn), are more commonly encountered than others. Fuel analysis should focus on identifying common sources of trace element contamination.

Warranty Summary

Cummins Inc. Engine Warranty covers malfunctions that are a result of defects in material or factory workmanship. Engine damage, service issues, and/or performance issues determined by Cummins Inc. to be caused by using fuels **not** conforming to appropriate specifications are **not** considered to be defects in material or workmanship and are **not** covered under Cummins Inc. engine warranty.

Fuel Sampling

Cummins Inc. recommends collection and analysis of fuel at periodic intervals to make sure the bulk fuel supply continues to meet all the requirements in Table 2 and provides adequate performance and protection of the engine and components. Cummins Inc. sampling frequency recommendations are applicable to all engines, all markets, and all fuel types; however, the practicality of maintaining as delivered fuel quality is most applicable to customers utilizing bulk fuel delivery and storage systems. The optimal frequency and level of analysis recommended will differ based on equipment type, user application, duty cycle, region, and environmental considerations. Given the large number of variables to consider, Cummins Inc. recommends all customers use the below schedule as best practice but work with the fuel supplier to custom tailor a fuel sampling and analysis program best suited for the situation.

Sampling Frequency and Methods

Several critical environmental, chemical, and handling events should be used to initiate fuel sampling and analysis. At a minimum, Cummins Inc. recommends fuel sampling and analysis be completed quarterly and/or after a full bulk inventory turn (cycle), whichever comes first. Additional triggers to consider include:

- New/adjusted fuel blends and changeovers (summer vs. winter fuel)
- Addition/changes to additive packages (injector deposit cleaner, cetane improvers, lubricity improvers, etc.)
- Change of fuel supplier
- New installations or customer operations
- Significant weather changes (rapid temperature swings, dust storms, heavy rain/snow, etc).

Fuel sampling techniques will depend on the location within the fuel supply chain to be sampled. Cummins Inc. recommends fuel samples be taken directly from vehicle/equipment at the inlet to suction side fuel filter. Contact the local Cummins Distributor for more information.

Analysis Recommendations

Cummins Inc. recommends fuel analysis be completed by an ISO/IEC 17025 certified laboratory, with experience in fuel analysis and capability to perform tests defined in Table 2. The quantity and type of test to be performed will determine the cost of the associated sampling. It is the customer's responsibility to understand and maintain the quality of the fuel used in their engine. A basic list of fuel sample analysis tests follows in Table 3.

Table 3, Recommended Fuel Sample Test Properties

Property/Item	**Test Priority (A or B) and Frequency**	**Purpose**
Anti-Knock Index	A-Every Sample	Engine Performance
Particulate Contamination	A – Every Sample	Prevent rapid erosion of fuel injection equipment – poor performance and component failure.

Table 3, Recommended Fuel Sample Test Properties		
Property/Item	**Test Priority (A or B) and Frequency**	**Purpose**
Water Content	A – Every Sample	Corrosion prevention, rapid fuel injection equipment wear out mechanisms.
Ethanol Content	A – Every Sample	Engine performance
Trace Contaminant Elements	A – Every Sample	Prevention of deposit formation, or exhaust aftertreatment component damage.
Sulfur Content	A – Every Sample	Prevent premature fuel injection equipment wear out. Fuel and lube oil system corrosion prevention. Exhaust aftertreatment component durability.
Simulated Distillation	A – Every Sample	Verify proper combustion, deposit mitigation
Biological Contamination	B - Quarterly	Corrosion prevention, premature fuel filter plugging.

Fuel Tank Care and Maintenance

Fuel tank cleaning is a major operation which requires complete draining of the fuel tank and should **only** be done by professionals. It is therefore carried out infrequently, normally on the schedule of several years, coinciding with (statutory) inspection and maintenance requirements. Good housekeeping can help extend the periods between fuel tank cleanings.

Water bottom measurements can be made at an appropriate time interval (via automatic gauging or regular tank dipping with water finding paste) and water can be removed when necessary. This is important since any water and sediment can be stirred up when the fuel tank is filled. Cummins Inc. recommends waiting a minimum of one hour per foot of fuel depth before dispensing fuel after delivery. If water and sediment are observed, additional settling time is one way of bringing the fuel back into specification.

It is virtually impossible to stop water from entering the supply chain; therefore, good housekeeping is essential. Hardware, fuel tanks, and pumping systems should be routinely inspected and maintained. Fuel should be checked periodically for contamination by water to make sure that there is no free water present in the fuel entering the engine.

Fuel Blending

Users of gasoline fuels should avoid blending in any substances that compromise fuel performance. Solvents, methanol, used lubricating oil and other contaminants **must not** be added to otherwise compliant fuels. Gasoline fuel should **only** be blended with ethanol in accordance with applicable fuel specifications.

Fuel Additives

⚠ CAUTION ⚠

Overuse of fuel additives can cause adverse effects such as fuel filter plugging, fuel system deposits, and reduced aftertreatment life.

In certain situations, fuel additives can be useful but in other cases, misuse can have detrimental effects. Cummins Inc. neither approves nor disapproves of the use of any fuel additive, fuel extender, fuel system modification, or the use of any device **not** manufactured or sold by Cummins Inc. or its subsidiaries. Engine damage, service issues, or performance problems that occur due to the use of these products are **not** considered a defect in workmanship or material as supplied by Cummins Inc. and will **not** be compensated for under the Cummins Inc. warranty. Cummins® engines are designed, developed, rated, and built to operate on commercially available gasoline fuel as described in Cummins Inc. Required Gasoline Fuel Specifications.

Cummins Inc. accepts no liability for engine damage resulting from the use of fuel additives that are **not** specifically approved. Consult your fuel supplier for guidance on additive use.

Fuel Filtration

Depending on the design of the system, on-engine fuel filtration may **not** be easily serviceable. Refer to OEM service information for guidance.